

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. A typical compound-Poisson configuration contains linearly many independent full-rank smoothing blocks; configurations without them have exponentially small probability. This regenerative decomposition gives global high-frequency bounds despite singular one-particle marks. Covariance comes from typical compound activity, and exact finite saddles yield the mixed microcanonical coefficient.

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1. ROUND-EIGHT RECONSTRUCTION: REGENERATIVE COMPOUND SMOOTHING AND EXACT SADDLE COEFFICIENTS

The continuous mark law of one particle is allowed to be singular. Smoothing is obtained from linearly many independent full-rank blocks in a typical compound-Poisson sample; the complement of this regenerative event is exponentially small. The covariance is computed from the typical compound law rather than from an exponentially rare fixed-particle sector.

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1.1. Admissible constraint marks. Let

$$X(v) = (v, |v|^2/2, \chi_1(v), \dots, \chi_d(v)) \in \mathbb{R}^{4+d}.$$

We call the constraint family *admissible* on a source chart K if the tilted second-moment matrices satisfy

$$\inf_{\lambda \in K} \inf_{|u|=1} \int (u \cdot X(v))^2 e^{\lambda \cdot X(v)} d\nu(v) > 0,$$

and if some product patch $U \subset (\text{supp } \nu)^{n_0}$ makes the sum map

$$S(v_1, \dots, v_{n_0}) = \sum_{j=1}^{n_0} X(v_j)$$

a submersion. These are explicit affine-span and transversality conditions; they exclude redundant constraints such as $\chi_1 \equiv 0$.

Lemma 1.1 (Finite submersion cover). *For an admissible family there are finitely many patches U_α and smooth cutoffs ζ_α such that the restricted pushforwards $S_*(\zeta_\alpha \nu^{\otimes n_0})$ have compactly supported $W^{k,1}$ densities, uniformly for every prescribed finite k and every source in K .*

Proof. On a full-rank patch choose a nonvanishing $(4+d)$ -minor and use the corresponding coordinates as transverse variables. Coarea gives a density; differentiating under the fiber integral gives $W^{k,1}$ bounds after the patch is shrunk away from the zero set of the minor. Compactness of the source chart and the support used by the saddle gives a finite cover. General velocity variations, not only radial variations in a coordinate plane, supply the momentum directions. $\square$

1.2. Good-block regeneration. Let N_μ be Poisson with mean μa_λ and split the first $n_0 \lfloor N_\mu/n_0 \rfloor$ marks into predetermined blocks. A block is *good* if it lies in one of the submersion patches, with the cutoff label chosen by an independent partition coin. Its probability is at least $p_* > 0$, uniformly on K .

Lemma 1.2 (Linearly many smooth blocks). *There are $c, C > 0$ such that*

$$\mathbb{P}_\lambda \{M_\mu < c\mu\} \leq C e^{-c\mu},$$

where M_μ is the number of good disjoint blocks. Conditional on the remaining marks and on $M_\mu = M$, the continuous sum law is a convolution of M uniformly $W^{k,1}$ densities with an arbitrary residual probability measure.

Proof. Poisson concentration gives $N_\mu \geq c_0 \mu$ outside an exponentially small event. Conditional on N_μ , the indicators of the predetermined good blocks are independent Bernoulli variables with success probability at least p_* . Chernoff's bound gives the first assertion. Independence of disjoint blocks gives the convolution decomposition; all singular marks are kept in the residual factor rather than declared smooth. $\square$

Theorem 1.3 (Global characteristic-function estimate). *For every k there are $C, c, R > 0$ such that the compound characteristic function satisfies*

$$|\Phi_{\mu,\lambda}(\xi)| \leq Ce^{-c\mu} + C_k(1 + |\xi|)^{-k}, \quad |\xi| \geq R,$$

uniformly for $\lambda \in K$. Lattice coordinates are restricted to one fundamental domain. Empty and low-particle sectors are contained explicitly in the term $Ce^{-c\mu}$.

Proof. On the event in Lemma 1.2, Fourier transformation gives

$$|\Phi(\xi \mid M)| \leq \prod_{j=1}^M |\hat{g}_j(\xi)|.$$

A $W^{k,1}$ density has $|\hat{g}_j(\xi)| \leq C_k|\xi|^{-k}$ at large frequency and, by nonlattice affine span, is uniformly less than one on every compact annulus away from zero. Since $M \geq c\mu$, the annular part is exponentially small and the large-frequency part has arbitrary polynomial decay. The complementary event, including every fixed small-particle sector, has probability $Ce^{-c\mu}$. $\square$

1.3. Typical-sector covariance. The normalized compound pressure is

$$Q(\lambda) = \int (e^{\lambda \cdot X(v)} - 1) d\nu(v).$$

Lemma 1.4 (Uniform covariance from compound activity). *For an admissible mark family,*

$$D^2Q(\lambda) = \int X(v)X(v)^T e^{\lambda \cdot X(v)} d\nu(v) \geq c_K I, \quad \lambda \in K.$$

Moreover, for the total mark C_{tot} ,

$$\mu^{-1} \text{Cov}_\lambda(C_{\text{tot}}) = D^2Q(\lambda).$$

Proof. The compound-Poisson logarithmic moment generating function is μQ . Differentiating twice proves both identities. The affine-span condition and compactness of K give the uniform positive lower bound. No conditioning on $N_\mu = n_0$ occurs. $\square$

1.4. Mixed lattice–continuous coefficient. Let $L_\mu \in \mathbb{Z}^r$ be the lattice constraints and $Y_\mu \in \mathbb{R}^s$ the continuous constraints. Let $(\lambda_\mu, \theta_\mu)$ be the unique exact finite saddle for the extensive target (ℓ_μ, y_μ) .

Theorem 1.5 (Exact finite-saddle coefficient). *Suppose the continuous shell is*

$$Y_\mu \in y_\mu + w_\mu B, \quad w_\mu = o(\sqrt{\mu}),$$

with w_μ measured in the extensive variable. Then

$$\mathbb{P}\{L_\mu = \ell_\mu, Y_\mu \in y_\mu + w_\mu B\} = e^{-\mu I_\mu} \frac{1}{(2\pi\mu)^{r/2}} \int_{w_\mu B/\sqrt{\mu}} \varphi_{\Sigma_{c,\mu}}(z) dz (1 + o(1)),$$

uniformly on compact admissible target charts. The formula includes the empty/small-sector exponential remainder.

Proof. Tilt at the exact finite saddle, so the Fourier phase has zero linear term. On the major arc use the third-order Taylor expansion and Lemma 1.4. The intermediate and high-frequency arcs are controlled by Theorem 1.3. Fourier inversion in *lattice* $\times$ *continuous* variables gives the coefficient. Undoing the tilt produces the exact exponent I_μ and no centering mismatch. $\square$

Theorem 1.6 (Microcanonical transfer and prepared LDP). *The exact coefficient theorem transfers bounded marked pressures, covariances, and finite-dimensional LDPs from the grand-canonical law to the extensive microcanonical shell. The transferred covariance is the Schur complement of D^2Q along the constrained directions.*

Proof. Apply Theorem 1.5 to the numerator with a bounded mark source and to the denominator without that source. The leading shell factors cancel uniformly. Differentiating the resulting pressure ratio at zero gives the conditional Schur complement. Exponential tightness and the finite-source Laplace principle give the prepared LDP. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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